Does DMD recover reproducible dynamics from deconvolved calcium-event data?

Initial validation plan before the full brain-state analysis

Planning checkpoint only. No DMD test or biological comparison has been run for this document. Implementation will begin only after this plan is reviewed. The workflow stops again after the initial validation report so that its results can be inspected before any all-recording analysis or mode tracking.

Project:	`ca_imaging_network_analysis/DMD`
Version:	1.1
Date:	16 July 2026
Parent plan:	`02_dmd_brain_state_analysis_plan.pdf`

Contents

1 Decision summary

Narrow decision to be made

The pilot asks whether at least one neuron-resolved observation and preprocessing arm yields a low-rank linear propagator that is (i) predictively useful on contiguous held-out data, (ii) numerically and bootstrap reproducible, (iii) distinguishable from smoothing and independent-neuron nulls, and (iv) recoverable under a small known-system simulation matched to empirical sparsity. It does *not* ask whether awake and unconscious states differ.

The closest direct precedent is Pachitariu et al., who applied a time-lagged, ridge-regularized DMD to neuronwise standardized, spike-deconvolved two-photon data after PCA, and deliberately used a 0.23-s lag to reduce sensitivity to short-timescale deconvolution artefacts [1]. We will reproduce the logic of that analysis, but not copy parameters that are incompatible with the present sampling rate and 300-frame target window.

The pilot has three possible conclusions:

PASS to expanded validation.
At least one arm passes the synthetic, predictive, reproducibility, and tracking-resolution gates. It becomes a candidate for validation on all recordings; no biological claim follows yet.

REVISE.
DMD is useful only after changing the signal, lag, rank, or window length. The parent plan is prospectively amended before further work.

STOP.
No arm is predictive and reproducible beyond the controls, or apparent success is reproduced by the smoothing/null pipeline. Intensive DMD and Grassmann tracking are not pursued with these data.

2 Questions and non-goals

2.1 Pilot questions

1. Can a ridge-DMD model predict held-out population activity better than persistence at physical horizons near 1 and 2 s?
2. Does the full population operator add information beyond independent per-PC autoregressions, or is predictability explained by marginal smoothing and calcium autocorrelation?
3. Are the leading eigenvalues and invariant subspace reproducible under moving-block resampling and deterministic neuron subsampling?
4. Can the estimator recover known real-relaxation versus rotational dynamics after sparse-event and calcium-like observation transformations?
5. Which observation arm—native deconvolution, binned events, stored smoothed events, or $\Delta F/F$—is empirically defensible for the next phase?

2.2 Explicit non-goals

Work excluded from this checkpoint

The pilot will not estimate awake–sleep or awake–anaesthesia effect sizes; train a state classifier; use state separation to choose preprocessing; implement SGOT or another Grassmann tracker; claim individual biological mode identity; analyze transitions; or run cohort-level inference. In-sample reconstruction is diagnostic and cannot by itself constitute validation.

3 What is copied from Pachitariu and what is revised

Pachitariu et al. standardized each neuron, projected activity into as many as 1,000 PCs, and fitted a ridge map from X_t to $X_{t+\Delta t}$, using a ridge penalty of 0.01 for two-photon data and $\Delta t \simeq 0.23$ s [1]. They compared discrete eigenvalue distributions with matched simulations rather than relying on a noisy inversion to a continuous generator. The audited reference is the public MouseLand/critical_init release v1.0; its version identifier will be recorded in the pilot manifest.

Reference-code audit and declared departures

Release v1.0 centers each neuron and divides by its root-mean-square deviation plus 10^{-3}; this is nearly, but not exactly, a textbook sample z-score. It applies an unnormalized Gram-matrix ridge of 0.01, so that number is not portable across sample counts or ranks. Its rotation code filters on $|\lambda| > 0.25$, whereas the associated figure caption describes a real-part threshold. The smoke test will reproduce the code definition; the report will also show the caption definition and label the discrepancy. The local-window implementation will use an SVD/method-of-snapshots solve rather than explicitly squaring the covariance condition number. These are recorded adaptations, not silent changes.

Prospective implementation amendment (version 1.1)

Before any empirical DMD fit was inspected, the implementation audit fixed four under-specified points. First, a $k = 4$ projector inside a $q = 4$ model is the identity and cannot measure mode stability; therefore only $q \geq 8$ models are eligible for the $k = 4$ subspace and tracking gates. Lower ranks remain eligible for prediction but can support only a conditional outcome. Second, "maximally time-spread" now means the global maximum of the minimum separation between integer window starts, with the lexicographically earliest tied solution. Third, development validation now mirrors deployment by fitting the first 80% and scoring the last 20% within each complete development window. Fourth, the approved raw-label windows remain primary, while Sz/Sc receive a seven-frame edge-trimmed sensitivity because the stored smoother can carry support across a nearby label transition. These corrections were frozen in the machine-readable configuration before examining a DMD result.

Element	Pachitariu-aligned choice	Required adaptation here
Observation	Per-neuron spike-deconvolved activity	This becomes the anchor, not a presumed winner; stored smoothed events, binned events, and $\Delta F/F$ are observation-model controls.
Scaling	Per-neuron centering and RMS scaling before PCA	Reproduce the release-v1.0 RMS-plus-10^{-3} rule, estimating both quantities from development data only. Never standardize each 300-frame window independently.
Spatial reduction	PCA to 1,000 dimensions	A near-literal 1,000-neuron sanity check is possible on `example_data`; a 300-frame window cannot identify 1,000 dimensions, so local fits use low fixed ranks.
Operator	Ridge regression in PC coordinates	Retain ridge DMD as the main estimator. Unregularized DMD is only a numerical benchmark.

Element	Pachitariu-aligned choice	Required adaptation here
Lag	0.23 s to reduce deconvolution artefacts	The closest lag is two frames, $2/7.65 = 0.261$ s. Four- and eight-frame lags test 0.523 and 1.046 s; one frame is reported only as an artefact-sensitive diagnostic.
Ridge penalty	Unnormalized $\alpha = 0.01$ for two-photon data	Retain the raw 0.01 fit as the precedent anchor, but select from a dimensionless, scale-relative training-only grid because effective shrinkage changes with dimension, sample count, and feature scale.
Fit duration	Long recordings at 22 Hz	Include a long-block diagnostic, then separately test the intended 300-frame (39.2-s) deployment window and a 1,500-frame (196.1-s) rescue window.
Endpoint	Eigenvalue complexity/rotations	Report this endpoint, but require forecast and subspace stability as additional feasibility evidence.

This is a faithful methodological adaptation rather than a literal replication. In particular, copying 1,000 PCs into 300 samples would guarantee an underdetermined local operator, and copying the absolute ridge penalty without auditing scaling would not reproduce the same regularization strength.

4 Development data and leakage control

4.1 Files and fixed selection rule

The smoke test uses `data/raw/example_data.mat`, which contains 1,000 neurons. Because that example is derived from the first sleep dataset, empirical evidence instead uses a separate sleep recording with adequate windows in both labels and the smallest-ROI anaesthesia recording, selected before any DMD result:

File	ROIs	Frames	Pilot role
`example_data.mat`	1,000	9,350	Code/numerical smoke test and near-literal high-dimensional Pachitariu configuration; not biological evidence.
`mouse02_sleep.mat`	6,574	18,700	Six disjoint awake and six disjoint NREM window candidates, using raw labels rather than `used_frame`.
`mouse07_ane.mat`	4,337	19,000	Six disjoint awake and six disjoint anaesthesia window candidates.

This keeps the first computation small while including both paradigms. The next phase, if approved, repeats the locked configuration on all ten recordings.

4.2 Window selection and label masking

Within each recording–state cell, choose six non-overlapping 300-frame windows that have one constant raw dataset label and remain inside one acquisition segment. Globally maximize the minimum separation between their integer start indices; among tied solutions, use the lexicographically earliest sequence. The project-specific `used_frame` variable is not an eligibility criterion. Split the six deterministically into three development and three untouched evaluation windows, alternating in time. This yields 12 untouched empirical evaluation windows.

State labels are permitted only to create approximately homogeneous windows and balance development data. During hyperparameter selection, their names are replaced by anonymous stratum identifiers, and no between-label performance or separability criterion is calculated.

For each recording and signal arm, neuron scaling and the common PCA basis are learned from the six development windows pooled with equal weight across the two anonymous strata. Hyperparameters use local blocked validation: within each development window, fit the first 80% and score the final contiguous 20%, then aggregate across the six anonymously. The PCA basis is then frozen for every evaluation window. This is essential: window-specific PCA would create artificial subspace motion and make later mode comparisons uninterpretable.

All neurons with finite, nonzero variance in the development data are eligible; no neuron is selected using state differences or DMD performance. The pilot will report the retained count and repeat the winning configuration on deterministic equal-N neuron subsets as a dimensional sensitivity.

5 Candidate observation and preprocessing arms

All arms preserve one row per neuron. No across-neuron averaging, binary threshold, or new Gaussian smoothing is applied. OASIS amplitudes are not treated as calibrated spike counts or spikes per second [2, 3].

ID	Signal construction	Scaling	Purpose
P	Native `spike_deconv` at 7.65 Hz	Train-only, code-matched centered RMS scaling	Pachitariu anchor; retains temporal resolution but is highly sparse.
B4	Non-overlapping four-frame sums of `spike_deconv`, divided by 0.523 s	Train-only centered RMS scaling	Reduces zero inflation without the 1.96-s stored smoothing. It is an event-mass-rate proxy, not calibrated firing rate.
Sz	Stored 15-frame/1.96-s `spike_smoothed` at 7.65 Hz	Train-only centered RMS scaling	Tests the signal used by the Kiyooka network analysis under Pachitariu-like scaling. No second smoothing is added.
Sc	Stored `spike_smoothed`	Recording/development mean-centering, no variance scaling	Restricted sensitivity matching the current parent plan; tests whether z-scoring creates or erases structure.
F	Native $\Delta F/F$	Train-only centered RMS scaling	Observation-model control: asks whether continuous calcium carries reproducible dynamics lost by deconvolution.

A nine-frame/1.176-s binned-event arm is deferred from the core screen because a 39.2-s window would contain only about 33 bins. It may be run as a rank-2 stress test if all higher-resolution event arms fail. If only $\Delta F/F$ succeeds, the conclusion is to change the primary observation model, not to declare the deconvolved signal validated. Calcium-aware latent-dynamics methods remain a later alternative if separate deconvolution and DMD are inadequate [4, 5].

Because the stored 15-frame smoother has an approximate seven-frame half-width, Sz/Sc are additionally rescored after trimming seven samples from each edge of every selected window. This is a declared boundary-support sensitivity; the untrimmed raw-label window remains the primary arm definition.

6 Model and minimal hyperparameter grid

For a fixed arm, common PCA basis, rank q, and lag ℓ, let $z_t \in \mathbb{R}^q$ be the PC score. Fit

$$z_{t+\ell} = Bz_t + \varepsilon_t, \qquad \widehat{B} = YX^\top \left(XX^\top + \alpha I\right)^{-1}. \tag{1}$$

The eigenvalues of $\widehat{B}$ are discrete-lag eigenvalues. The pilot will not convert them to a continuous generator unless a value is well separated from zero and the complex-log branch is unambiguous.

For deployment fits, define the scale $s_X = \mathrm{tr}(XX^\top)/q$ and write $\alpha = \eta s_X$. Thus η is the ridge penalty relative to the mean eigenvalue of the predictor Gram matrix. The literal Pachitariu smoke-test fit instead uses the original unnormalized $\alpha = 0.01$.

6.1 Three fit tiers

1. **Precedent check.** On `example_data`, run arm P with the release-v1.0 neuronwise RMS scaling, the maximum numerically supported PCA dimension up to 1,000, unnormalized $\alpha = 0.01$, and a two-frame lag. Reproduce the discrete-eigenvalue plane and rotations-per-tenfold-attenuation summary of Pachitariu et al. This checks implementation fidelity, not biological validity.
2. **Deployment check.** In the 300-frame development/evaluation windows, use a common recording-level basis and low ranks. Native arms use $q \in \{2, 4, 8, 16\}$, additionally capped at no more than one retained dimension per ten eligible training snapshot pairs. B4 contains only 75 bins and about 60 fitting bins, so it uses $q \in \{2, 4\}$.
3. **Long-block diagnostic.** In each of the four recording–label cells, select the 1,500-frame block centered in the longest eligible constant-label bout, without reference to DMD output. Refit at $q \in \{4, 8, 16, 32\}$ using blocked training-only selection and a final contiguous 20% evaluation segment. This is a prespecified rescue test that distinguishes estimator failure from insufficient 39.2-s support; it is not an opportunity to select a favourable biological contrast.

Native-arm lags are $\ell \in \{1, 2, 4, 8\}$ frames, corresponding to 0.131, 0.261, 0.523, and 1.046 s; two frames is the precedent anchor and one frame is diagnostic. For Sz/Sc, a 15-frame (1.961-s) lag additionally tests the scale of the stored smoothing kernel. B4 uses one-, two-, and four-bin lags.

The raw precedent value $\alpha = 0.01$ is always shown. On development data only, deployment selection uses

$$\eta \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}.$$

Both η and the resulting raw α are recorded. Exact DMD ($\eta = 0$) is a conditioning diagnostic, not an eligible winner when it is unstable.

7 Held-out empirical validation

7.1 Blocked prediction

In each untouched 300-frame evaluation window, estimate the local operator from the first 80% (240 frames, or 60 B4 bins) and evaluate on the final contiguous 20%. No shuffled pair, acquisition break, state boundary, or artificial bootstrap-block join may enter $X \to Y$.

Rolling-origin forecasts reinitialize from the observed state and assess the nearest available horizons to 0.26, 1.05, and 2.09 s. For baseline b, define

$$\mathrm{Skill}_b(h) = 1 - \frac{\sum_t \| z_{t+h} - \hat{z}^{\mathrm{DMD}}_{t+h} \|_2^2}{\sum_t \| z_{t+h} - \hat{z}^{(b)}_{t+h} \|_2^2}. \tag{2}$$

Positive skill means that DMD improves on that baseline. Report variance-weighted and equal-PC versions, normalized RMSE, correlation, and the fraction of non-finite or explosive forecasts.

Required baselines are:

1. training mean;
2. persistence, $\hat{z}_{t+h} = z_t$; and
3. an independent ridge AR model for each PC at the same lag and horizon.

Beating persistence alone is insufficient for the smoothed signal. Full-DMD gain over diagonal AR is the evidence for cross-component population dynamics. Its absence does not make stable real relaxation spectra meaningless, but it forbids a claim of coordinated/rotational modes.

7.2 Numerical and spectral diagnostics

For every fit, retain the effective rank, ridge scale, normal-matrix condition number, residual, spectral radius, non-finite flag, and forecast blow-up flag. Plot the discrete eigenvalue plane directly. Following Pachitariu et al., report the number of rotations per tenfold attenuation for sufficiently nonzero complex eigenvalues, but do not interpret a frequency with fewer than three cycles in the window or a branch-ambiguous logarithm.

7.3 Bootstrap and neuron-subset reproducibility

With the PCA basis fixed, perform 100 moving-block resamples per evaluation window. Block length is at least 15 native frames (about 2 s) and may increase with the empirical autocorrelation time. Snapshot pairs are formed only inside blocks. Group a conjugate pair as one real invariant subspace.

For a development-only reference operator with $q \geq 8$, select $k = 4$ real invariant dimensions by mean training modal-coefficient energy, never splitting a conjugate pair. Match every evaluation and bootstrap fit back to those fixed reference groups. Compare a bootstrap projector $P^{(b)}$ with its reference P using

$$S_{\mathrm{sub}} = \frac{1}{k} \operatorname{tr}\left(P P^{(b)}\right) = \frac{1}{k} \sum_{j=1}^{k} \cos^2 \theta_j. \tag{3}$$

Also report matched discrete-eigenvalue spread and forecast uncertainty. Repeat the chosen arm on two deterministic, equal-size neuron subsets to show whether the conclusion depends on a particular population sample.

Use normalized projector distance $d_P = \sqrt{\max(0, 1 - S_{\mathrm{sub}})}$. If fewer than four nontrivial matched dimensions exist, or if $k = q$, subspace stability is uninformative and the corresponding gate is not eligible to pass.

To determine whether future tracking has adequate resolution, compute

$$R_{\mathrm{track}} = \frac{\mathrm{median}(\text{within-window bootstrap projector distance})}{\mathrm{median}(\text{between-window same-label projector distance})}. \tag{4}$$

If estimation noise is not clearly smaller than ordinary between-window change, a visually smooth Grassmann trajectory would not be trustworthy.

8 Positive controls and negative controls

8.1 Lightweight known-system recovery

Run 20 deterministic seeds for each of two stable six-dimensional systems: one with real relaxation modes and one with a known conjugate rotational pair. Project each latent system into approximately 500 neuronal observables and test four observation levels, following the causal ordering of the precedent's observation simulation where applicable:

1. continuous Gaussian observations (software upper bound);
2. rectified latent rates followed by sparse Poisson/event observations, matched to the empirical zero fraction;
3. the same events after a 15-frame Gaussian smoothing approximation; and
4. a 0.25-s exponential calcium convolution with matched shot noise.

This is intentionally smaller than the end-to-end simulator in the parent plan. OASIS re-estimation and switching/mode-birth simulations belong to expanded validation if this first screen passes. The lightweight test must nevertheless show when sparsity or smoothing changes the recovered eigenspectrum.

8.2 Empirical negative controls

For each evaluation window, generate 100 neuronwise circular-shift controls inside the same continuous, constant-label bout. They preserve each neuron's marginal distribution and autocorrelation approximately while disrupting synchronous population relationships. Also simulate independent stationary event/calcium traces with empirical rates and the same smoothing.

DMD may predict these controls because marginal autocorrelation is genuine. The null expectation is instead that full-DMD gain over diagonal AR and reproducible rotational/coupled subspaces disappear. A smoothed trace is not validated merely because it is easy to forecast.

9 Predeclared screening gates

These are feasibility thresholds, not biological hypothesis-test thresholds. They are frozen before running the pilot and reported even when they fail.

Gate 1 — geometry and numerical health

There must be zero snapshot pairs across an acquisition boundary, raw-label change, or bootstrap-block join. At least 95% of evaluation fits must produce finite eigenvalues and finite forecasts with complete provenance.

Gate 2 — known-system recovery

Across the sparse/calcium-like synthetic observations, at least 80% of seeds must correctly distinguish real-relaxation from rotational systems; median S_{sub} must be at least 0.75; identifiable frequency/decay error must be no more than 25%; and held-out forecast skill must be positive.

Gate 3 — empirical predictive validity

At least one arm must have positive median skill over persistence near 1 s, with a moving-block-bootstrap lower confidence bound above zero in at least three of the four recording–state cells. It must not show systematic forecast failure at about 2 s, and fewer than 5% of fits may be non-finite or explosive.

Gate 4 — empirical reproducibility

Median S_{sub} must be at least 0.80 in at least three of the four recording–state cells, and the qualitative spectral class must agree across the two deterministic neuron subsets.

Gate 5 — tracking resolution

$R_{\text{track}} < 0.5$ in both development recordings. If a 300-frame arm fails but its prespecified 1,500-frame version passes, the outcome is REVISE the window rather than PASS the original sliding-window design.

Coordinated-mode interpretation qualifier

A claim of multivariate coupling or a reproducible rotational mode additionally requires full-DMD gain over diagonal AR, or the relevant rotational statistic, to exceed the 95th percentile of the circular-shift null. Failure of this qualifier permits only a conditional statement about stable marginal/relaxation spectra; it does not satisfy a coordinated-mode claim.

10 Decision rule and the review pause

Decision	Evidence pattern	Authorized next step after review
PASS	At least one arm passes Gates 1–5. Any coordinated-mode claim also passes the null qualifier.	Freeze that candidate and propose expanded validation on all recordings. Do not start tracking automatically.
CONDITIONAL	Prediction and stability pass, but diagonal/null evidence is weak, or only a 1,500-frame/$\Delta F/F$ configuration passes.	Revise signal, window, or interpretation; limit the endpoint to reproducible decay/eigenspectrum or whole-subspace summaries.
FAIL	Synthetic event recovery fails, no empirical arm beats persistence, or subspaces are dominated by estimation/null variation.	Stop intensive DMD; report the identifiability bound and retain static rate/covariance or a calcium-aware latent-model alternative.

The implementation phase ends after generating the artifacts below. The user reviews the evidence and explicitly decides whether to expand, revise, or stop.

11 Required implementation artifacts

All future DMD-specific files remain inside `DMD/`:

- thin entry point: `DMD/scripts/00_initial_dmd_validation.py`;
- reusable implementation and tests: `DMD/src/` and `DMD/tests/`;
- immutable run output: `DMD/results/03_initial_validation/`; and
- result report: `DMD/documents/04_dmd_initial_validation_results.pdf`.

The result directory must include a machine-readable configuration, software versions and seeds, selected-window manifest, per-fit metrics, bootstrap metrics, null distributions, and explicit failure records. No DMD implementation will be placed in the repository's tutorial scripts/, src/funcnet, or general results/ directories.

Required report figures, in order, are:

1. signal distributions, zero fraction, autocorrelation, and representative all-neuron/PC traces for each arm;
2. the Pachitariu-style discrete-eigenvalue plane and rotation statistic;
3. forecast skill versus physical horizon, arm, lag, and rank, with both persistence and diagonal-AR baselines;
4. representative held-out PC predictions;
5. bootstrap eigenvalue clouds and invariant-subspace stability;
6. known-system recovery across observation levels;
7. observed versus circular-shift/null multivariate evidence; and
8. a traffic-light table showing every gate, exception, and final decision.

12 Relationship to the parent analysis plan

This document is a prospective pilot supplement, not a silent rewrite of 02_dmd_brain_state_analysis_plan.pdf. The parent document currently names native spike_smoothed, unscaled centering, and a one-frame lag as primary choices. Those choices are now treated as unresolved until this pilot is reviewed. Raut et al. remain a benchmark for continuous widefield-$\Delta F/F$ DMD, not the precedent for DMD on sparse per-neuron events [6].

If the pilot passes, its locked signal, scaling, lag, rank, estimator, and window will be written as a versioned amendment to the parent plan before state labels are used for biological comparison. If the pilot fails, the negative result is the scientifically correct stopping point.

References

[1] Marius Pachitariu, Lin Zhong, Alexa Gracias, Amanda Minisi, Crystall Lopez, and Carsen Stringer. A critical initialization for biological neural networks. *Nature*, 2026. doi: 10.1038/s41586-026-10528-1. URL https://doi.org/10.1038/s41586-026-10528-1.

[2] Johannes Friedrich, Pengcheng Zhou, and Liam Paninski. Fast online deconvolution of calcium imaging data. *PLOS Computational Biology*, 13(3):e1005423, 2017. doi: 10.1371/journal.pcbi.1005423. URL https://doi.org/10.1371/journal.pcbi.1005423.

[3] Daiki Kiyooka, Ikumi Oomoto, Jun Kitazono, Yoshihito Saito, Midori Kobayashi, Chie Matsubara, Kenta Kobayashi, Masanori Murayama, and Masafumi Oizumi. Single-cell resolution functional networks during unconsciousness are segregated into spatially intermixed modules. *Cell Reports*, 45(2):116902, 2026. doi: 10.1016/j.celrep.2025.116902. URL https://doi.org/10.1016/j.celrep.2025.116902.

[4] Ziqiang Wei, Bei-Jung Lin, Tsai-Wen Chen, Kayvon Daie, Karel Svoboda, and Shaul Druckmann. A comparison of neuronal population dynamics measured with calcium imaging and electrophysiology. *PLOS Computational Biology*, 16(9):e1008198, 2020. doi: 10.1371/journal.pcbi.1008198. URL https://doi.org/10.1371/journal.pcbi.1008198.

[5] Tze Hui Koh, William E. Bishop, Takashi Kawashima, Brian B. Jeon, Ranjani Srinivasan, Yu Mu, Ziqiang Wei, Sandra J. Kuhlman, Misha B. Ahrens, Steven M. Chase, et al. Dimensionality reduction of calcium-imaged neuronal population activity. *Nature Computational Science*, 3:71–85, 2023. doi: 10.1038/s43588-022-00390-2. URL https://doi.org/10.1038/s43588-022-00390-2.

[6] Ryan V. Raut, Zachary P. Rosenthal, Xiaodan Wang, Hanyang Miao, Zhanqi Zhang, Jin-Moo Lee, Marcus E. Raichle, Adam Q. Bauer, Steven L. Brunton, Bingni W. Brunton, and J. Nathan Kutz. Arousal as a universal embedding for spatiotemporal brain dynamics. *Nature*, 647(8089):454–461, 2025. doi: 10.1038/s41586-025-09544-4. URL https://doi.org/10.1038/s41586-025-09544-4.